Interlaboratory Evaluation of Radiated and Conducted EMC Measurements Using Portable Calibrated Comb Generators

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***Abstract*—This paper presents an interlaboratory evaluation of radiated and conducted electromagnetic compatibility (EMC) measurements using portable calibrated comb generators as repeatable reference sources. The same reference generators were employed in two independent EMC laboratories under equivalent test conditions to assess measurement consistency and identify potential sources of variability associated with instrumentation and test facilities. Radiated emissions were evaluated in semi-anechoic chambers at a 3 m test distance over the frequency range from 30 MHz to 1 GHz, while conducted emissions were assessed from 150 kHz to 30 MHz using line impedance stabilization networks (LISNs). Quasi-peak radiated results were compared at 21 common frequencies, producing root-mean-square errors (RMSEs) of 5.4 dB for horizontal polarization and 10.2 dB for vertical polarization. Conducted results were compared at 12 common frequencies, producing RMSEs of 3.0 dB for the line conductor and 2.7 dB for the neutral conductor. The results demonstrate that portable calibrated comb generators provide a practical and repeatable approach for EMC interlaboratory comparison exercises, enabling the evaluation of measurement consistency across independent facilities. The methodology was additionally implemented as part of a final-year undergraduate Electromagnetic Compatibility course, providing practical experience in EMC measurements, traceability, and interlaboratory comparison techniques.**

***Index Terms*—electromagnetic compatibility, EMC testing, interlaboratory comparison, radiated emissions, conducted emissions, comb generator, measurement consistency, measurement reproducibility.**

I. Introduction

Electromagnetic compatibility (EMC) testing depends on measurement results that are both reliable and traceable. These requirements are especially important when measurements are used for equipment characterization, compliance assessment, product development, and verification activities. The accuracy of radiated and conducted emission measurements is influenced not only by the equipment under test, but also by the complete measurement chain, including antennas, receivers, line impedance stabilization networks (LISNs), cables, correction factors, test-site characteristics, and data-processing procedures [1], [2], [5].

For this reason, international standards place strong emphasis on laboratory competence, measurement traceability, and quality assurance. Accredited laboratories are expected to periodically verify the validity of their measurements through activities such as interlaboratory comparisons and proficiency-testing exercises [3], [4]. These activities provide valuable information about the consistency of measurement results obtained under nominally equivalent conditions.

Radiated and conducted emission measurements are particularly sensitive to differences in test setups. Variations in chamber performance, antenna calibration, receiver configuration, cable losses, and environmental conditions can contribute to measurement deviations even when the same standards and procedures are followed [1], [2], [5]. Evaluating the magnitude of these differences is therefore useful for understanding the overall performance of EMC measurement facilities.

One practical challenge in interlaboratory comparisons is the selection of a suitable reference source. Commercial electronic equipment often exhibits variability associated with operating modes, software configuration, loading conditions, or manufacturing tolerances. Portable calibrated comb generators offer an attractive alternative because they produce stable harmonic spectra over broad frequency ranges and can be transported between laboratories while maintaining repeatable measurement conditions.

This work presents an interlaboratory comparison performed between Laboratory A and Laboratory B using portable calibrated comb generators as common reference sources for radiated and conducted emission measurements. The same reference sources were measured independently in both laboratories, allowing the resulting spectra to be directly compared. The objective was not to perform a formal proficiency-testing exercise, but rather to evaluate the level of agreement obtained under realistic operating conditions and to identify potential sources of measurement variability.

II. Objective

The primary objective of this work is to evaluate the consistency of radiated and conducted emission measurements obtained at two independent EMC laboratories using portable calibrated comb generators as common reference sources.

A secondary objective is to compare the spectral amplitudes, harmonic locations, and overall spectral-envelope characteristics measured under equivalent test conditions. For radiated emissions, the comparison covers measurements performed in semi-anechoic chambers from 30 MHz to 1 GHz at a test distance of 3 m. For conducted emissions, the analysis covers the range from 150 kHz to 30 MHz using LISNs.

The working hypothesis is that the use of stable and repeatable reference sources should allow the principal spectral characteristics to be reproduced with reasonable agreement across independent facilities, despite differences in instrumentation, chamber characteristics, and measurement implementation.

Because this preliminary campaign does not include complete laboratory-specific uncertainty budgets or formal proficiency-testing criteria, the results are evaluated through direct comparison of the measured spectra and selected quantitative indicators rather than through a pass/fail assessment. The study therefore focuses on the practical evaluation of measurement consistency and on the identification of potential sources of variability associated with EMC test facilities.

III. Experimental Setup

The interlaboratory comparison was organized into two complementary measurement campaigns: radiated emissions and conducted emissions. In both cases, portable calibrated comb generators were used as common reference sources and transported between the participating laboratories. This approach minimized variability associated with the test object and allowed the measurement systems themselves to be evaluated under realistic operating conditions.

The comparison was performed independently at Laboratory A and Laboratory B. Although both laboratories followed equivalent measurement procedures, each facility used its own instrumentation, chamber infrastructure, calibration data, and operational practices. The objective was not to reproduce identical measurement environments, but rather to evaluate the level of consistency obtained when the same reference sources were measured in different EMC facilities.

A. Radiated Emission Measurement Setup

Radiated emission measurements were performed in semi-anechoic chambers using a Com-Power CGO-505 comb generator, serial number 301360, as the common reference source. The generator produces harmonic components spaced at 5 MHz intervals over a frequency range from 5 MHz to 1.5 GHz and can be easily transported between laboratories while maintaining stable operating conditions.

For both measurement campaigns, the generator was positioned on a nonconductive table 0.8 m above the ground plane and located 3 m from the receiving antenna. Measurements covered the frequency range from 30 MHz to 1 GHz using a resolution bandwidth of 120 kHz. A BIC-BBA 9106-01 biconical antenna was used from 30 MHz to 230 MHz, while an LPD-ULSP 9143-01 log-periodic antenna was used from 230 MHz to 1 GHz. Horizontal and vertical antenna polarizations were evaluated using equivalent source orientations. Broadband scans were acquired using a Peak detector and selected frequencies were subsequently evaluated using Quasi-Peak measurements.

TABLE I
Test Parameters

Parameter	Radiated	Conducted
Frequency range	30 MHz–1 GHz	150 kHz–30 MHz
Detector	Peak/Quasi-Peak	Quasi-Peak
RBW	120 kHz	9 kHz
Test distance	3 m	Not applicable
Reference source	CGO-505	CGC-255E
Environment	Semi-anechoic chamber	LISN setup

B. Conducted Emission Measurement Setup

Conducted emission measurements were performed using a Com-Power CGC-255E comb generator, serial number 311402, as the common conducted-emission reference source. The generator produces stable harmonic components from 50 kHz to 115 MHz and was transported between laboratories following the same comparison methodology adopted for radiated measurements.

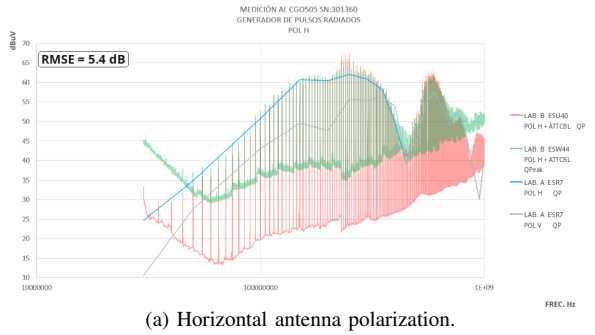
The comparison covered the frequency range from 150 kHz to 30 MHz using a resolution bandwidth of 9 kHz. At both laboratories, the generator output was connected through a Rohde & Schwarz ENV216 LISN and disturbance voltages were measured independently on the line and neutral conductors in dBμV. At Laboratory A, background measurements were first acquired with the generator powered off, followed by active measurements with the generator operating normally. The generator was positioned 0.4 m above the metallic reference plane.

C. Comparison Methodology

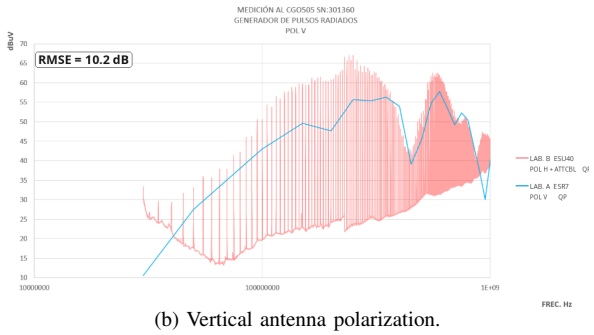
The comparison was based on direct evaluation of the spectra obtained independently at both laboratories. Particular attention was given to the identification of common harmonic components, spectral-envelope characteristics, amplitude differences, and overall measurement consistency. Radiated-emission measurements were evaluated separately for horizontal and vertical polarizations, while conducted-emission measurements were analyzed independently for the line and neutral conductors.

All spectra were represented using common frequency and amplitude scales to facilitate visual comparison. In addition, selected Quasi-Peak measurements obtained at common frequencies were compared using root-mean-square error (RMSE) as a descriptive indicator of agreement between laboratories. The reported RMSE values are used to summarize the observed differences and are not intended to represent measurement uncertainty or formal proficiency-testing acceptance criteria.

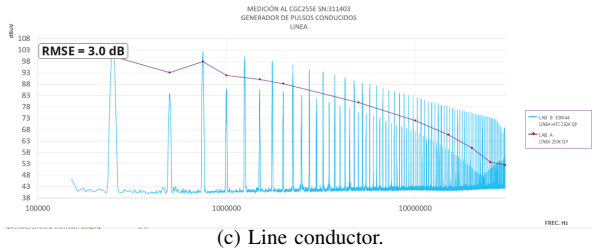
For target frequencies not explicitly present in the 30-kHz Laboratory B sweeps, the closest available spectral sample was selected. The interlaboratory residual was defined as $\Delta = L_B - L_A$.



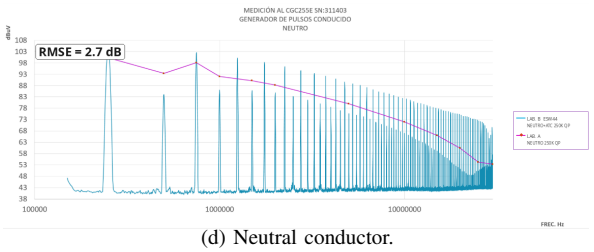
(a) Horizontal antenna polarization.



(b) Vertical antenna polarization.



(c) Line conductor.



(d) Neutral conductor.

Fig. 1. Comparison of radiated- and conducted-emission measurements obtained by Laboratory A and Laboratory B.

IV. Results

A. Radiated Emission Results

The radiated-emission spectra obtained with the CGO-505 reference source are shown in Fig. 1 for horizontal and vertical antenna polarizations. The spectra measured at both laboratories exhibit the same principal harmonic structure across the evaluated frequency range, confirming the suitability of the portable reference source for interlaboratory comparison.

For horizontal polarization, the spectral envelopes show good overall correspondence between approximately 100 MHz and 400 MHz. Both laboratories identify the same dominant spectral features and similar amplitude trends throughout most of the measurement range. Above approximately 500 MHz, the differences between the measured levels become more noticeable, with increasing divergence toward the upper end of the spectrum. Despite these differences, the principal harmonic components remain clearly identifiable in both facilities.

A similar behavior is observed for vertical polarization. Common harmonic locations and overall spectral trends are reproduced by both measurement systems, although larger amplitude deviations are observed compared to the horizontal configuration. This result highlights the greater sensitivity of radiated-emission measurements to factors such as antenna polarization, chamber characteristics, source positioning, and measurement-system implementation.

The quantitative comparison of the Quasi-Peak measurements yielded an RMSE of 5.4 dB for horizontal polarization and 10.2 dB for vertical polarization. These values indicate that the horizontal-polarization measurements exhibited closer agreement between laboratories than the vertical-polarization measurements under the evaluated conditions.

Figure 2 shows the frequency-dependent residuals. For horizontal polarization, 15 of the 21 differences remain within ± 5 dB, with the largest deviation occurring near 950 MHz. The vertical residuals exhibit greater dispersion and a stronger positive bias; only 9 of 21 points remain within ± 5 dB, with the largest difference occurring at 30 MHz.

B. Conducted Emission Results

The conducted-emission spectra obtained with the CGC-255E reference source showed a high degree of similarity between laboratories for both line and neutral conductors. The common harmonic structure generated by the source was consistently reproduced throughout the frequency range from 150 kHz to 30 MHz.

For the line conductor, both laboratories measured nearly identical spectral envelopes, including the principal amplitude maxima and the overall decay trend toward higher frequencies. The neutral-conductor results exhibited a similar behavior, with comparable harmonic amplitudes and spectral shapes across the entire measurement range.

The quantitative comparison yielded RMSE values of 3.0 dB for the line conductor and 2.7 dB for the neutral conductor. These values are lower than those obtained for the radiated-emission measurements, indicating closer agreement between laboratories for the conducted measurements.

From an EMC perspective, this result is consistent with the more controlled measurement environment provided by the LISN-based conducted-emission setup. Unlike radiated measurements, which depend on test-site characteristics, antenna factors, and chamber performance, conducted measurements are largely referenced to a defined impedance environment, reducing the influence of laboratory-specific effects.

V. Discussion

The results demonstrate that portable calibrated comb generators can be effectively used as common reference sources for interlaboratory EMC comparison exercises. The same reference sources were transported and measured independently

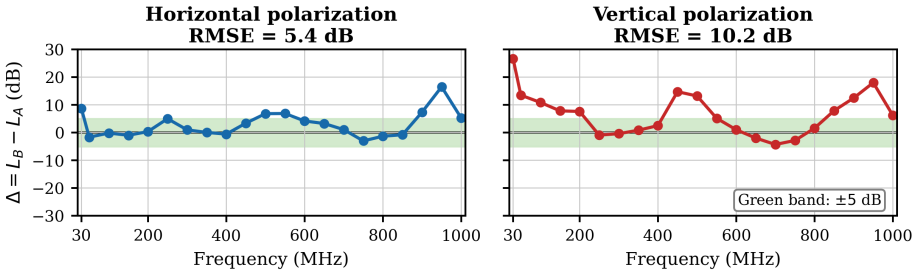


Fig. 2. Radiated interlaboratory error versus frequency, $\Delta = L_B - L_A$; shaded region: ± 5 dB.

at Laboratory A and Laboratory B, allowing the influence of the measurement facilities and instrumentation to be evaluated while minimizing variability associated with the test object.

For conducted emissions, both laboratories reproduced highly similar spectral envelopes and harmonic structures. The obtained RMSE values of 3.0 dB and 2.7 dB for the line and neutral conductors indicate a close level of agreement under independent measurement conditions. This behavior is consistent with the relatively controlled measurement environment provided by LISN-based testing, where the influence of chamber characteristics and radiated coupling mechanisms is minimized.

The radiated-emission measurements exhibited larger differences, particularly for vertical polarization. Although the harmonic structure and overall spectral trends were consistently reproduced, the amplitude deviations increased at higher frequencies and produced RMSE values of 5.4 dB for horizontal polarization and 10.2 dB for vertical polarization. Such differences are expected to be more sensitive to antenna factors, chamber performance, source positioning, cable losses, and receiver configuration.

From an EMC perspective, the observed results should be interpreted in the context of the variability typically encountered in emission measurements. Civil EMC standards such as CISPR 16-4-2 establish measurement instrumentation uncertainty levels on the order of 3 dB to 5 dB for many conducted- and radiated-emission measurement configurations [2]. Under this context, the RMSE values obtained for conducted emissions and horizontal-polarization radiated emissions are of the same order as the uncertainties commonly encountered in EMC practice. The larger deviation observed for vertical polarization suggests that laboratory-dependent effects become more pronounced under that measurement configuration and deserve further investigation.

An important outcome of this work is the demonstration that portable calibrated comb generators provide a practical methodology for evaluating measurement consistency across independent EMC facilities. Unlike conventional electronic equipment, whose emissions may vary with operating modes, software configuration, environmental conditions, or manufacturing tolerances, the reference sources employed in this study produced stable and repeatable harmonic spectra that could be reproduced at different laboratories with minimal operational complexity.

Several limitations should nevertheless be acknowledged. The reported RMSE values are intended as descriptive indicators of agreement between laboratories and should not be interpreted as measurement uncertainty values or as formal conformity-assessment criteria. A more rigorous comparison would require complete laboratory-specific uncertainty budgets and uncertainty-normalized compatibility analysis in accordance with CISPR 16-4-2. Future work should also include additional laboratories and repeated measurement campaigns to establish a broader statistical basis for evaluating interlaboratory consistency.

VI. Conclusions

This work presented an interlaboratory evaluation of radiated and conducted EMC measurements using portable calibrated comb generators as common reference sources. Transporting the same sources between Laboratory A and Laboratory B enabled direct comparison of complete measurement chains while minimizing variability associated with the test object.

Both laboratories identified the principal harmonic structure of the sources. Conducted-emission RMSE values were 3.0 dB for line and 2.7 dB for neutral, while radiated-emission values were 5.4 dB for horizontal and 10.2 dB for vertical polarization. The conducted and horizontal-radiated differences are of the same order as uncertainty levels commonly encountered in EMC practice. Although this work does not constitute a formal uncertainty analysis or proficiency-testing exercise, it demonstrates that portable calibrated comb generators provide a practical and repeatable methodology for evaluating measurement consistency across independent EMC facilities.

References

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